Rain-Aware Multibeam Satellite Scheduling via Sequential Reinforcement Learning and Graph-Based Shielded Execution

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***Abstract*—Adaptive beam scheduling in multibeam satellite networks is challenging under geographically nonuniform traffic, rain-induced channel degradation, and co-channel interference from full frequency reuse. Heuristic schedulers are sensitive to coupled traffic–channel dynamics, while direct reinforcement learning over joint multibeam assignments suffers from combinatorial action growth and weak structural credit assignment. This paper proposes a shielded sequential reinforcement learning framework for rain-aware multibeam satellite scheduling. Each physical scheduling interval is reformulated as a sequence of conditional beam-selection decisions, where the policy observes the system state and a beam-assignment mask encoding previously selected cells. This factorization exposes inter-beam spatial dependency while preserving simultaneous physical beam activation. A PPO agent is trained in a physics-informed storm-aware synthetic benchmark with diurnal traffic, queue dynamics, geographically nonuniform demand, and ITU-R-inspired rain attenuation. At deployment, an inference-time graph-based shield projects the raw beam sequence onto a conflict-free active set. Experiments on a 63-cell benchmark show that PPO-Tier achieves 19.66 units/step throughput with zero shielded interference, outperforming the best queue-based greedy baseline at 16.97 units/step. Results confirm the benefits of sequential factorization, tier-aware reward shaping, and graph-based shielded execution.**

***Index Terms*—Multibeam satellite networks, beam scheduling, reinforcement learning, Proximal Policy Optimization, rain attenuation, interference-aware scheduling, cognitive satellite communications.**

I. INTRODUCTION

High-throughput satellite (HTS) systems increasingly rely on aggressive frequency reuse, flexible beam illumination, and beam hopping to support geographically heterogeneous service demand. In such systems, beam scheduling is no longer a static resource allocation problem, but a dynamic decision-making task that must continuously adapt to spatially uneven traffic, time-varying queue backlogs, fluctuating channel conditions, and interference constraints. This challenge becomes more pronounced in tropical and subtropical regions, where rain attenuation can significantly distort the instantaneous service capability of Ka/Ku-band satellite links and create strongly nonstationary operating conditions.

Recent studies have shown that machine learning and reinforcement learning can improve radio resource management and beam hopping decisions in multibeam satellite networks beyond conventional heuristics [1]–[3]. Learning-based optimization has been investigated for dynamic beam illumination, non-orthogonal access, interference-aware beam hopping, and service-driven scheduling in both GEO and NGSO settings [4]–[8]. More broadly, recent work on terrestrial and non-terrestrial network integration has highlighted machine learning as a key enabler for adaptive decision making in next-generation communication systems [9]. However, despite this progress, most existing studies still rely on simplified or slowly varying channel assumptions, focus on static or quasi-static impairment patterns, or emphasize throughput maximization without jointly addressing dynamic rain-induced service degradation, geographically asymmetric demand, inter-beam dependency, and deployment-time interference feasibility.

Rain attenuation is one of the dominant impairments in Ka/Ku-band satellite systems and remains a key challenge for adaptive satellite communications [10], [11]. Since attenuation depends on both rain intensity and the spatio-temporal evolution of rain cells, a scheduler that ignores storm-induced channel variation may waste beam resources on severely faded regions while neglecting nearby cells with better effective service opportunities. Although recent adaptive GEO beam hopping studies have considered nonuniform user distributions and flexible beam allocation [12], [13], the interaction between moving rain-induced degradation and reinforcement learning-based beam scheduling remains insufficiently explored. This gap is particularly relevant for tropical service regions, where the scheduler must react not only to nonuniform demand, but also to time-varying rain-affected capacity.

A second challenge arises from the action formulation. In a multibeam scheduler, several beams must be assigned within the same physical scheduling interval. Directly selecting multiple beams over many candidate cells leads to a combinatorial action space and weak structural credit assignment. In common MultiDiscrete policy parameterizations, beam decisions are often represented by multiple categorical action heads conditioned on the same observation. Such a formulation makes it difficult for the policy to capture spatial dependency among simultaneously activated beams. When a selected beam pattern causes interference, duplication, or wasted service opportunity, it is also difficult for the policy optimizer to identify which individual beam decision is responsible for the degraded reward.

A third challenge concerns deployment-time feasibility under full frequency reuse. Simultaneously activating adjacent or strongly coupled beams may cause severe co-channel interference, and the deployed beam pattern should satisfy scheduling-layer compatibility constraints at runtime. Existing optimization-based and physical-layer approaches can mitigate interference, but they often require richer channel information, iterative computation, joint power control, or precoding operations that may be undesirable for a lightweight scheduling

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layer [2], [5], [14]. In contrast, a graph-based scheduling abstraction provides a low-complexity way to represent beam compatibility constraints. Under this abstraction, the scheduler should avoid activating conflicting cells in the same physical scheduling interval, regardless of policy stochasticity or transient model uncertainty.

Motivated by these challenges, this paper proposes a shielded sequential reinforcement learning framework for rain-aware beam scheduling in multibeam satellite networks. Instead of predicting all beam assignments jointly, we reformulate each physical scheduling interval as a sequence of conditional beam-selection decisions. This sequential factorization allows the policy to condition each new beam selection on previously assigned beams within the same macro scheduling step, thereby exposing inter-beam spatial dependency to the learning agent and reducing the effective action complexity. A Proximal Policy Optimization (PPO) agent is trained in a storm-aware environment that combines geographically nonuniform demand, diurnal traffic dynamics, queue evolution, and a physically motivated rain attenuation process inspired by established satellite propagation models [10], [11], [15]. To support deployment under full frequency reuse, we further introduce an inference-time shield that projects the raw beam sequence onto a conflict-free active set defined by a scheduling-layer interference graph. The shield is a deployment-time feasibility layer rather than a constrained-RL training mechanism; it guarantees conflict-free activation only with respect to the adopted graph abstraction and does not replace physical-layer SINR modeling, precoding, or power allocation.

From a cognitive networking perspective, the proposed scheduler follows a perception–reasoning–execution cycle. The perception layer observes queue backlogs, demand arrivals, rain-dependent capacity factors, geographical information, time-of-day context, and the partial beam-assignment mask. The reasoning layer uses a PPO policy with conditional sequential factorization to construct a multibeam schedule. The execution layer then applies a deterministic graph-based shield when strict one-hop interference avoidance is required. This separation allows the policy to learn adaptive scheduling behavior from data while preserving deployment-time feasibility under the adopted scheduling-layer model.

The proposed framework is evaluated on a 63-cell benchmark derived from geographically asymmetric demand conditions and multiple operating scenarios, including no-rain, storm-affected, and reduced-load regimes. The results show that the sequential PPO scheduler improves useful throughput, packet loss, hotspot satisfaction, and conflict avoidance compared with heuristic and MultiDiscrete PPO baselines. In addition, the comparison between shielded and unshielded execution is used to distinguish between interference-aware behavior learned by the policy itself and the deterministic feasibility guarantee provided by the shield. Tier-aware reward shaping further improves service protection for high-demand urban hotspots while preserving strong demand-based fairness measured by a satisfaction-based fairness index [16], [17]. These findings suggest that combining sequential policy factorization with graph-based shielded execution is a promising design principle for cognitive beam scheduling in dynamic satellite communication networks.

The main contributions of this work are summarized as follows:

- We formulate rain-aware multibeam satellite scheduling as a conditional sequential decision problem. Instead of predicting all beam targets simultaneously, each physical scheduling interval is decomposed into a sequence of beam-selection sub-decisions conditioned on the previously selected cells. This formulation reduces the effective action complexity and exposes inter-beam spatial dependency to the policy.
- We introduce a deterministic inference-time shield that projects the raw beam sequence onto a conflict-free active set under a scheduling-layer interference graph. The shield provides deployment-time feasibility with respect to the adopted graph abstraction, while the unshielded results are used to assess how much interference-aware behavior is learned by the policy itself.
- We develop a physics-informed storm-aware scheduling benchmark that combines geographically nonuniform demand, diurnal traffic variation, queue dynamics, and ITU-R-inspired rain attenuation. The benchmark is designed to evaluate adaptive scheduling under structured nonstationarity rather than to reproduce a full meteorological digital twin.
- Using no-rain, storm-affected, and reduced-load scenarios, we show that the proposed PPO-Tier scheduler improves useful throughput, packet loss, hotspot satisfaction, and conflict avoidance compared with heuristic and MultiDiscrete PPO baselines. The ablation results further isolate the effects of sequential factorization, tier-aware reward shaping, and shielded execution.

The remainder of the paper is organized as follows. Section II reviews related work on learning-based beam scheduling, rain-aware satellite resource management, and constraint-aware scheduling. Section III presents the system model and problem formulation. Section IV describes the proposed shielded sequential reinforcement learning framework. Section V presents the experimental setup and results. Section VI concludes the paper and discusses future research directions.

II. Related Work

A. Learning-Based Satellite Beam Scheduling

Beam hopping and dynamic beam illumination are key mechanisms for improving spectrum utilization and traffic adaptability in high-throughput satellite (HTS) systems. Early reinforcement learning-based studies showed that learning-based beam hopping can outperform conventional heuristics under dynamic traffic conditions [1], [2]. Subsequent works extended this direction to more complex resource management problems, including NOMA-assisted beam hopping [4], dynamic beam illumination with selective precoding [5], multi-satellite beam hopping with load balancing and interference avoidance [6], interference-aware beam hopping [7], service-driven beam hopping [8], integrated satellite-terrestrial resource allocation [18], and sequential dynamic resource allocation [19].

These studies demonstrate the value of learning-enabled satellite scheduling, but several limitations remain. First, many existing methods formulate multibeam scheduling as a simultaneous or joint action-selection problem. Although this formulation is natural from a physical scheduling perspective, it can lead to a large combinatorial action space when multiple beams must be assigned over many candidate cells. In common MultiDiscrete policy parameterizations, different beam decisions are generated by separate categorical action heads conditioned on the same observation, which makes inter-beam dependency difficult to capture explicitly. Second, several learning-based beam hopping studies focus mainly on traffic adaptation or average throughput improvement, while the combined effects of rain-induced channel degradation, geographically asymmetric demand, queue evolution, and deployment-time interference feasibility remain less explored. These gaps motivate a conditional sequential policy formulation in which a multibeam schedule is generated through a sequence of dependent beam-selection decisions, while still being executed as a simultaneous beam pattern at the physical scheduling interval.

B. Rain-Aware and Interference-Constrained Satellite Resource Management

Traffic demand in satellite coverage areas is often highly nonuniform, especially when a GEO satellite serves both dense urban regions and sparse rural areas. Recent studies on adaptive multibeam GEO scheduling have shown that flexible beam allocation can improve utilization under nonuniform ground-user distributions [12], [13]. However, these works do not explicitly integrate moving rain-induced degradation into a reinforcement learning-based scheduling policy.

Rain attenuation is a dominant impairment in high-frequency satellite communications, particularly for Ka/Ku-band links [10], [11]. Space-time stochastic rain models and hybrid rain-cell models have shown that rain attenuation is inherently spatio-temporal rather than a purely scalar link impairment [10], [11]. Nevertheless, most learning-based beam scheduling studies still use static, slowly varying, or simplified channel assumptions. This paper bridges these directions by embedding a lightweight ITU-R-inspired storm-aware attenuation process into the scheduling environment. The resulting benchmark is not intended to reproduce a full meteorological digital twin; rather, it provides structured and geographically meaningful rain-induced nonstationarity for evaluating adaptive scheduling policies.

Interference is another critical issue under full frequency reuse. Existing satellite interference mitigation approaches often rely on optimization-based scheduling, precoding, power allocation, or detailed physical-layer channel information [2], [5], [14]. While effective, such methods may require richer channel state information or higher computational complexity than desired for a lightweight scheduling layer. In contrast, this work adopts a graph-based scheduling-layer abstraction, where an edge between two cells indicates that simultaneous activation should be avoided. This abstraction does not replace physical-layer SINR modeling, precoding, or power allocation. Instead, it provides a conservative compatibility rule that can be enforced efficiently during deployment.

C. Constraint-Aware Execution and Positioning of This Work

Constraint handling in reinforcement learning has been widely studied for safety-critical and resource-constrained decision-making problems. Representative approaches include constrained policy optimization, Lagrangian-based methods, penalty-based formulations, risk-sensitive learning, and shielding-based techniques [20]–[22]. These methods are conceptually related to the need for feasible beam activation, but the objective of this paper is different. We do not formulate satellite beam scheduling as a constrained reinforcement learning problem during training, nor do we aim to provide general safe-RL guarantees. Instead, we focus on a scheduling-layer deployment requirement: the final activated beam set should be conflict-free under the adopted graph-based interference abstraction.

The proposed framework therefore separates policy learning from deployment-time feasibility enforcement. The reinforcement learning policy is trained to generate a raw sequence of beam selections in the original action space. At inference time, a deterministic graph-based shield projects this raw sequence onto a conflict-free active set when strict one-hop interference avoidance is required. This design keeps the learning problem lightweight, allows the policy to explore and learn interference-aware behavior from reward feedback, and provides a simple feasibility guarantee with respect to the adopted conflict graph.

This positioning differs from prior learning-based beam hopping studies in three aspects. First, the proposed method uses conditional sequential policy factorization to expose inter-beam spatial dependency to the learning agent. Second, it evaluates scheduling under structured rain-induced capacity degradation, geographically asymmetric demand, and queue dynamics. Third, it distinguishes between learned interference-aware behavior and deterministic deployment-time feasibility by comparing shielded and unshielded execution. Together, these elements provide a cognitive scheduling framework for rain-aware multibeam satellite systems operating under full frequency reuse.

III. System Model and Problem Formulation

This section presents the multibeam satellite scheduling model, including the service coverage, traffic and queue dynamics, storm-aware attenuation, scheduling-layer interference constraints, and sequential decision formulation. The model is designed as a scheduling-layer benchmark for evaluating adaptive beam selection under geographically asymmetric demand, rain-induced capacity degradation, and graph-based interference feasibility. The overall interaction among traffic arrivals, queue evolution, beam hopping, and service outcomes is illustrated in Fig. 1.

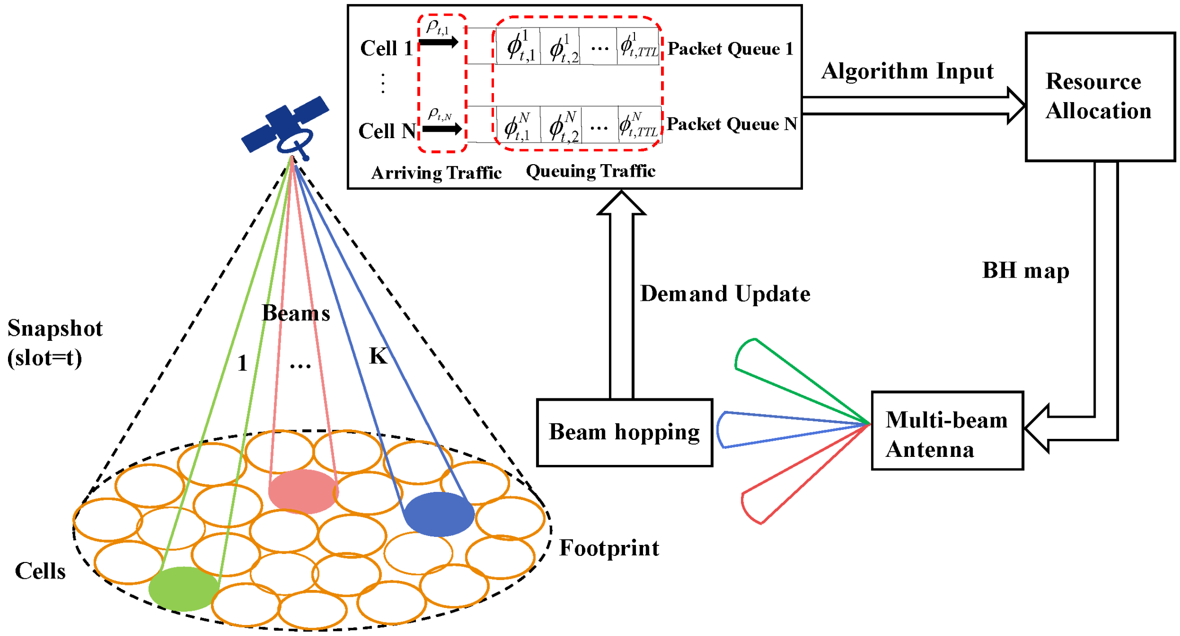

Fig. 1. System overview of the considered multibeam satellite scheduling framework. Traffic arrives at geographically distributed cells, accumulates in packet queues, and is served through beam hopping and resource allocation decisions under a limited beam budget.

A. Multibeam Service, Traffic, and Queue Model

We consider a multibeam satellite system serving a territory partitioned into N service cells. In the benchmark used in this paper, $N = 63$, corresponding to 63 geographically distinct service regions. At each physical scheduling interval, the satellite can activate at most K beams simultaneously, where $K = 4$. The system operates under full frequency reuse; hence, co-channel interference between nearby or strongly coupled simultaneously illuminated cells must be controlled [5]–[7].

Let $\mathcal{V} = \{1, 2, \ldots, N\}$ denote the set of service cells. The scheduling-layer interference relation is modeled by an undirected conflict graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $(i, j) \in \mathcal{E}$ means that cells i and j are incompatible for simultaneous activation within the adopted scheduling abstraction. This graph is used as a lightweight compatibility model for full-frequency-reuse beam scheduling. It does not replace physical-layer SINR modeling, precoding, or power allocation; rather, it provides a conservative rule for deciding whether a candidate active beam set is feasible at the scheduling layer.

To represent geographically asymmetric service importance, each cell is assigned a tier $\tau_i \in \{1, 2, 3\}$, where Tier 1 denotes high-demand hotspots, Tier 2 medium-demand regions, and Tier 3 lower-demand or remote regions. The tier index is not a hard constraint and does not directly determine whether a cell can be served. It is used only in the reward design to encourage service protection for high-priority cells [12], [13], [17].

A physical scheduling interval is referred to as a macro-step. During one macro-step, the physical state is fixed while the agent generates K beam selections over K internal sub-steps. After the full raw sequence is formed, the deployed active set is determined, and service delivery, queue update, packet loss, and storm motion are evaluated once at the macro-step level. This convention preserves simultaneous beam activation at the physical layer while allowing the policy to construct the multibeam schedule sequentially.

Traffic demand is spatially nonuniform and time-varying. For each cell i, a normalized base demand is defined by

$$L_{\text{base},i} = \frac{\text{Pop}_i}{\max_j \text{Pop}_j} L_{\max}, \tag{1}$$

where Pop_i is the population-associated service weight and $L_{\max}$ is the maximum base demand scaling factor. The exogenous demand arrival at macro-step t is

$$D_i(t) = \min(L_{\text{base},i} M(t) \eta_i(t), D_{\max}), \tag{2}$$

where $\eta_i(t)$ is a clipped Gaussian perturbation centered at one and $D_{\max}$ limits unrealistically large arrivals. The diurnal multiplier is

$$M(t) = 1 + 0.5 \sin\left(2\pi \frac{h(t) - h_{\text{peak}}}{24}\right), \tag{3}$$

where $h(t) \in [0, 24)$ is the hour-of-day associated with macro-step t, and h_{peak} denotes the daily peak-demand hour.

Each cell maintains a queue of unserved traffic. Let $q_i(t)$ be the queue backlog and let $\mathcal{A}(t) \subseteq \mathcal{V}$ be the deployed active cell set at macro-step t, with $|\mathcal{A}(t)| \leq K$. The set $\mathcal{A}(t)$ may be identical to the raw beam sequence proposed by the policy or may be obtained after applying the inference-time graph-based shield described in Section IV. If cell i is activated, its served load is

$$s_i(t) = \begin{cases} \min(q_i(t) + D_i(t), \mu_i C_i(t)), & i \in \mathcal{A}(t), \\ 0, & i \notin \mathcal{A}(t), \end{cases} \tag{4}$$

where μ_i is the nominal clear-sky service capacity and $C_i(t) \in [0,1]$ is the rain-dependent capacity factor. The queue evolves as

$$q_i(t+1) = \min\{Q_{\max}, q_i(t) + D_i(t) - s_i(t)\}, \quad (5)$$

and the overflow loss is

$$\ell_i(t) = \max\{q_i(t) + D_i(t) - s_i(t) - Q_{\max}, 0\}. \quad (6)$$

A waiting-time variable is also maintained for cells whose normalized queue ratio exceeds a critical threshold. It is used as both an observation feature and a delay-aware component in the reward.

B. Physics-Informed Storm-Aware Attenuation Model

Rain attenuation is a major impairment in high-frequency satellite links and has been widely studied using space-time stochastic rain fields and rain-cell models [10], [11]. In this work, we use a lightweight ITU-R-inspired storm-aware attenuation model. The environment should be interpreted as a physics-informed synthetic benchmark rather than a full meteorological digital twin. Its purpose is to provide structured nonstationarity and geographically meaningful rain-induced service degradation for evaluating adaptive scheduling policies.

At the beginning of an episode, a storm may be activated according to a Bernoulli random variable:

$$z_{\text{storm}} \sim \text{Bernoulli}(p_{\text{storm}}), \quad (7)$$

where p_{storm} is the storm activation probability. If $z_{\text{storm}} = 0$, clear-sky capacity is assumed and $C_i(t) = 1$ for all cells unless otherwise specified. If $z_{\text{storm}} = 1$, the storm is described by a center location, heading, storm-center translation speed, spatial spread, and peak rain intensity. The translation speed controls the movement of the storm center across the footprint and should not be interpreted as wind speed or tropical cyclone category. Rain severity is instead governed by the peak intensity and the induced rain-rate field.

Let $\mathbf{c}(t)$ denote the storm center. Its motion is modeled as

$$\mathbf{c}(t+1) = \mathbf{c}(t) + \mathbf{v}_{\text{step}}, \quad (8)$$

where $\mathbf{v}_{\text{step}}$ is derived from the storm-center translation speed, heading, and macro-step duration. For cell i, let $d_i(t)$ be the distance from the cell center to the storm center. A normalized rain intensity is generated using a Gaussian spatial profile:

$$I_i(t) = I_{\max} \exp\left(-\frac{d_i^2(t)}{\sigma^2}\right), \quad (9)$$

where $I_{\max}$ is the peak rain intensity and σ controls the storm spread. The intensity is converted into rain rate $R_i(t)$ and then mapped to attenuation using an ITU-R-inspired pipeline. The specific attenuation follows Recommendation ITU-R P.838:

$$\gamma_{R,i}(t) = kR_i^{\alpha}(t), \quad (10)$$

where k and α are frequency- and polarization-dependent coefficients [23]. The path attenuation is approximated using an effective slant-path length inspired by ITU-R P.618 and P.839 [24], [25]:

$$A_i(t) = \gamma_{R,i}(t) L_{\text{eff},i}. \quad (11)$$

The attenuation is converted into a normalized capacity factor:

$$C_i(t) = \text{clip}\left(1 - \frac{A_i(t)}{A_{\text{margin}}}, 0, 1\right), \quad (12)$$

where A_{margin} is the assumed link margin. This factor modulates the service rate in (4). Numerical settings of the attenuation model are reported in the experimental section.

C. Scheduling-Layer Interference Constraint and Sequential Decision Formulation

Under full frequency reuse, a deployed schedule should avoid incompatible adjacent or strongly coupled cell activations. A deployed active set $\mathcal{A}(t)$ is feasible if

$$(i,j) \notin \mathcal{E}, \quad \forall i,j \in \mathcal{A}(t),\ i \neq j. \quad (13)$$

Equivalently, $\mathcal{A}(t)$ must belong to the family of independent sets of the conflict graph:

$$\mathcal{A}(t) \in \mathcal{I}(\mathcal{G}). \quad (14)$$

This binary conflict abstraction is a conservative scheduling-layer model. It provides a hard compatibility rule for deployed beam patterns, but does not model continuous SINR degradation or physical-layer interference cancellation. In the proposed framework, this constraint is enforced at inference time by the graph-based shield described in Section IV.

Directly selecting all K beam targets in one step creates a combinatorial joint action space and weak structural credit assignment. A simultaneous ordered action has size N^K, while even duplicate-free unordered beam sets grow as $\binom{N}{K}$. We therefore formulate each macro-step as K internal sub-steps. At sub-step m, the agent selects one candidate cell conditioned on the physical state and the previously selected cells within the same macro-step.

Let x_t denote the physical state at macro-step t, and let $\mathbf{a}_{t,1:m-1}$ denote the ordered partial assignment before sub-step m. The conditional policy is factorized as

$$\pi(\mathbf{a}_t \mid x_t) = \prod_{m=1}^{K} \pi(a_{t,m} \mid x_t, \mathbf{a}_{t,1:m-1}), \quad (15)$$

where $\mathbf{a}_t = (a_{t,1}, \ldots, a_{t,K})$ is the raw beam sequence generated by the policy. This factorization is a policy-generation mechanism for constructing a simultaneous K-beam schedule; it does not imply that beams are physically activated at different times.

The raw sequence $\mathbf{a}_t$ is converted into a deployed active set by an execution mapping:

$$\mathcal{A}(t) = \mathcal{P}_{\mathcal{G}}(\mathbf{a}_t, q(t)), \quad (16)$$

where $\mathcal{P}_{\mathcal{G}}(\cdot)$ denotes either the identity mapping in unshielded evaluation or the graph-based projection applied by the inference-time shield. This notation separates policy generation from deployment-time feasibility enforcement and allows both shielded and unshielded policies to be evaluated.

At each sub-step, the observation includes per-cell queue, capacity, arrival, waiting-time, and geographic features; a sinusoidal time-of-day encoding; a beam-assigned mask; and a sub-step indicator. The beam-assigned mask allows the policy to condition each new beam decision on the partial schedule already constructed, enabling it to learn spatial dependency among beams without requiring a monolithic joint distribution over all combinations.

D. Reward and Optimization Objective

The scheduler is optimized to balance useful throughput, queue stability, packet loss mitigation, hotspot protection, and beam-use efficiency. The macro-step reward is

$$r_t = \frac{1}{\lambda}\Big(W_{\rm th}\Phi_t - W_{\rm loss}L_t - W_{\rm waste}B_t - W_{\rm q}\sum_{i=1}^{N}\hat{q}_i^2(t) - \sum_{i=1}^{N}P_{{\rm wait},i}(t)\Big), \tag{17}$$

where λ is a scaling factor, Φ_t is weighted useful throughput, L_t is weighted overflow loss, B_t is the wasted beam count, $\hat{q}_i(t) = q_i(t)/Q_{\max}$, and $P_{{\rm wait},i}(t)$ is a nonlinear waiting penalty.

The weighted useful throughput is

$$\Phi_t = \sum_{i\in\mathcal{A}(t)} \omega_i(t)s_i(t), \tag{18}$$

where $\omega_i(t)$ is a tier-aware urgency weight. A generic form is

$$\omega_i(t) = \beta_{\tau_i}\left(1+\kappa u_i(t)\right), \tag{19}$$

with tier coefficient β_{τ_i}, urgency gain κ, and normalized queue urgency

$$u_i(t) = {\rm clip}\left(\frac{\hat{q}_i(t) - q_{\rm low}}{q_{\rm high} - q_{\rm low}}, 0, 1\right). \tag{20}$$

The weighted overflow loss is

$$L_t = \sum_{i=1}^{N}\tilde{\omega}_i(t)\ell_i(t), \tag{21}$$

where $\tilde{\omega}_i(t)$ is a tier-aware loss weight.

The wasted-beam term B_t counts beam selections that do not contribute useful service because of duplication, graph-based conflict removal, or activation of cells with no effective service opportunity. This term discourages beam patterns that consume scheduling opportunities without increasing useful throughput. The waiting penalty $P_{{\rm wait},i}(t)$ discourages persistent starvation by increasing when a cell remains in a high-queue state for multiple macro-steps.

The learning objective is

$$\max_{\pi}\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1}\gamma^t r_t\right]. \tag{22}$$

The deployed schedule is subject to

$$|\mathcal{A}(t)| \le K, \tag{23}$$

$$\mathcal{A}(t) \in \mathcal{I}(\mathcal{G}), \tag{24}$$

$$0 \le C_i(t) \le 1, \quad \forall i \in \mathcal{V}. \tag{25}$$

The independent-set constraint in (24) is enforced only at deployment by the inference-time shield. It is not treated as a constrained-RL objective during training. The next section describes the proposed shielded sequential reinforcement learning framework.

IV. Proposed Shielded Sequential Reinforcement Learning Framework

This section presents the proposed framework for rain-aware multibeam scheduling. The design combines conditional sequential policy generation, PPO-based policy optimization, and an inference-time graph-based shield. The key principle is to separate *policy learning* from *deployment-time feasibility enforcement*. During training, the agent explores the raw action space and learns the consequences of interference, duplication, and wasted service through reward feedback. During inference, the graph-based shield can be applied to project the raw beam sequence onto a conflict-free active set under the adopted scheduling-layer interference graph. This guarantee is defined only with respect to the graph-based scheduling abstraction in Section III, rather than a full physical-layer SINR, precoding, or power-allocation model.

A. Sequential Policy Generation

A direct multibeam scheduling formulation requires the agent to select all K target cells simultaneously. If each beam can be assigned to one among N cells, the ordered joint action space grows as N^K, while even unordered duplicate-free beam sets still grow as $\binom{N}{K}$. In common MultiDiscrete policy parameterizations, the joint action is often represented by K categorical heads conditioned on the same observation. This formulation is convenient, but it does not explicitly expose the dependency between beam choices within the same physical scheduling interval. As a result, interference, duplicated assignments, or wasted service opportunities can lead to weak structural credit assignment.

To address this issue, the proposed scheduler generates one physical multibeam schedule through a sequence of K conditional decisions. At sub-step m, the policy selects one candidate cell while observing the physical state and the partial assignment already generated in sub-steps $1,\ldots,m-1$. Following the formulation in (15), the raw beam sequence is factorized as a product of conditional decisions. This factorization is a policy-generation mechanism only; it does not imply that beams are physically activated at different times. The physical state is frozen during the K internal sub-steps, and all selected beams are evaluated jointly at the end of the macro-step.

Let $\mathbf{b}_{t,m-1} \in \{0,1\}^N$ denote the beam-assigned mask after the first $m-1$ sub-decisions in macro-step t, where $b_{t,m-1,i} = 1$ if cell i has already been selected. The sub-step observation can be written as

$$o_{t,m} = \left[x_t,\, \mathbf{b}_{t,m-1},\, \mathbf{e}_m\right], \tag{26}$$

where x_t denotes the physical macro-step state and $\mathbf{e}_m$ is a one-hot indicator of the current sub-step. The effective action at each sub-step is a single discrete choice over N cells:

$$a_{t,m} \sim \pi_\theta(\cdot \mid o_{t,m}). \tag{27}$$

After selecting $a_{t,m}$, the mask is updated as

$$b_{t,m,i} = \max\left(b_{t,m-1,i}, \mathbb{I}[i = a_{t,m}]\right), \quad i \in \mathcal{V}. \tag{28}$$

Algorithm 1 Shielded Sequential PPO Scheduling Framework

Require: Policy π_θ, value function V_ϕ, conflict graph $\mathcal{G}$, beam budget K

Ensure: Trained scheduling policy π_θ

for each training episode **do**
  Initialize queues $\mathbf{q}(0)$, traffic state, and storm state
  for each macro-step t **do**
    Observe physical macro-state x_t
    Initialize raw beam sequence $\mathbf{a}_t \leftarrow [\,]$
    Initialize beam-assigned mask $\mathbf{b}_{t,0} \leftarrow \mathbf{0}$
    for $m = 1$ to K **do**
      Construct sub-step observation
$$o_{t,m} \leftarrow [x_t, \mathbf{b}_{t,m-1}, \mathbf{e}_m]$$
      Sample one beam target
$$a_{t,m} \sim \pi_\theta(\cdot \mid o_{t,m})$$
      Append $a_{t,m}$ to $\mathbf{a}_t$
      Update beam-assigned mask $\mathbf{b}_{t,m}$
      Store sub-step transition information
    end for
    Obtain deployed active set
$$\mathcal{A}(t) \leftarrow \mathcal{P}_{\mathcal{G}}(\mathbf{a}_t, \mathbf{q}(t))$$
    Evaluate service $s_i(t)$, packet loss $\ell_i(t)$, and reward r_t
    Update queues $\mathbf{q}(t+1)$ and storm state
    Assign the macro-step reward r_t to the stored sub-step transitions
  end for
  Update π_θ and V_ϕ using PPO
end for

This mask makes the partial beam pattern directly visible to the neural policy and helps it learn interference-aware and duplication-aware scheduling behavior without imposing a hard action mask during training.

Algorithm 1 summarizes the complete training and execution flow of the proposed framework. Within each macro-step, the physical state is frozen while the policy generates K conditional beam selections. Only after the raw sequence is completed is the deployed active set obtained through the execution mapping $\mathcal{P}_{\mathcal{G}}(\cdot)$. Service delivery, queue evolution, packet loss, and reward computation are then evaluated once at the macro-step level. This procedure preserves simultaneous physical beam activation while allowing the policy to learn a sequential factorization of the multibeam scheduling decision.

B. Observation, Reward, and PPO Training

At each internal sub-step, the physical state x_t includes per-cell queue backlog, rain-dependent capacity factor, current demand arrival, waiting-time indicator, and normalized geographical coordinates. These features are concatenated with a sinusoidal time-of-day encoding, a binary beam-assigned mask, and a sub-step indicator. With five per-cell physical features, two time-encoding features, an N-dimensional assignment mask, and a K-dimensional sub-step indicator, the observation dimension is

$$d_{\text{obs}} = 5N + 2 + N + K. \tag{29}$$

For $N = 63$ and $K = 4$, this gives $d_{\text{obs}} = 384$.

The reward follows the formulation in Section III. It combines weighted useful throughput, weighted packet loss, beam waste, queue congestion, and waiting-time penalties. Tier-aware weights increase the benefit of serving high-priority cells and the penalty for overflow in those cells, while queue and waiting penalties discourage persistent congestion. The evaluation metrics are kept separate from the reward so that the reported performance indicators are not directly optimized as standalone objectives.

The sequential scheduler is optimized using Proximal Policy Optimization (PPO) [15]. The policy and value functions are represented by multilayer perceptrons that take the structured sub-step observation as input. Each transition corresponds to one internal sub-step, while the service outcome and reward are computed only after all K selections have been generated. For the intermediate sub-steps, the environment stores the partial decisions and delays the macro-step service evaluation until the full raw sequence is available.

The PPO update maximizes the clipped surrogate objective

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t,\, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{30}$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\theta_{\text{old}}}(a_t \mid o_t)} \tag{31}$$

is the probability ratio and $\hat{A}_t$ is the advantage estimate. In implementation, the clipped policy loss is combined with value-function regression and entropy regularization, following standard PPO.

Two implementation choices are used to stabilize learning. First, the rollout horizon is synchronized with a complete simulated daily cycle so that each update observes both low-load and peak-load periods. This reduces phase-dependent bias caused by the diurnal traffic pattern. Second, the storm activation probability is increased gradually during training. The policy first learns demand-aware scheduling and spatial separation under clear-sky or low-storm-probability conditions, and is then exposed to more frequent rain-induced degradation. This curriculum controls how often storm-affected episodes appear; it is not a tropical cyclone category or wind-speed severity scale.

C. Inference-Time Graph-Based Shield

Although the policy is trained in the raw action space, the deployed beam set may be required to satisfy the conflict-free condition in (13). The inference-time graph-based shield converts the raw beam sequence into a feasible active set under the adopted conflict graph when strict one-hop interference avoidance is required. Using the notation introduced in Section III, the deployed active set is

$$\mathcal{A}(t) = \mathcal{P}_{\mathcal{G}}(\mathbf{a}_t, q(t)), \tag{32}$$

Algorithm 2 Inference-Time Graph-Based Shield

Require: Raw beam sequence $\mathbf{a}$, queue vector $\mathbf{q}$, conflict graph $\mathcal{G}$

Ensure: Conflict-free active set $\mathcal{A}$

$\mathcal{A} \leftarrow \textsc{RemoveDuplicates}(\mathbf{a})$
for all $c \in \mathcal{A}$ **do**
  $\mathcal{N}(c) \leftarrow \textsc{Neighbors}(c, \mathcal{G})$
  $\mathrm{cf}[c] \leftarrow |\mathcal{N}(c) \cap \mathcal{A}|$
end for
while $\max_{c \in \mathcal{A}} \mathrm{cf}[c] > 0$ **do**
  $c^- \leftarrow \arg\max_{c \in \mathcal{A}} (\mathrm{cf}[c], -q[c])$
  $\mathcal{A} \leftarrow \mathcal{A} \setminus \{c^-\}$
  for all $u \in \mathcal{N}(c^-) \cap \mathcal{A}$ **do**
    $\mathrm{cf}[u] \leftarrow \mathrm{cf}[u] - 1$
  end for
  delete $\mathrm{cf}[c^-]$
end while
return $\mathcal{A}$

where $\mathcal{P}_{\mathcal{G}}$ denotes the graph-based projection operator. In unshielded evaluation, $\mathcal{P}_{\mathcal{G}}$ reduces to the identity mapping after duplicate handling, which allows us to assess how much interference-aware behavior is learned by the policy itself.

The shield first removes duplicated cell selections. It then computes the number of conflicts for each selected cell. While conflicts remain, the shield removes the cell with the largest number of conflicts; ties are broken by removing the cell with lower queue urgency. This rule preserves more urgent cells while eliminating the most conflict-inducing selections. The procedure is summarized in Algorithm 2.

Algorithm 2 returns an independent set of the conflict graph:

$$\mathcal{P}_{\mathcal{G}}(\mathbf{a}_t, q(t)) \in \mathcal{I}(\mathcal{G}). \tag{33}$$

This guarantee is with respect to the scheduling-layer graph model, not a full physical-layer SINR model. Since the shield operates only on the at-most-K cells proposed within one macro-step, its worst-case computational complexity is $O(K^2)$.

The shield does not replace pruned beams with new candidates. Therefore, shielded execution may activate fewer than K beams in a macro-step. This avoids turning the shield into a second scheduler and preserves the separation between policy generation and feasibility enforcement. The proposed shield is not a constrained-RL training method; it acts only as a deterministic deployment-time projection layer.

D. Computational Complexity and Design Rationale

At each macro-step, the PPO policy generates K discrete decisions sequentially, so neural-network inference scales linearly with K. If C_π denotes the cost of one policy forward pass, the policy-generation cost per macro-step is $O(KC_\pi)$. The shield checks conflicts only among the selected cells, resulting in $O(K^2)$ complexity. Thus, the overall scheduling-layer execution cost is

$$O(KC_\pi + K^2). \tag{34}$$

This design is lightweight compared with optimization-based methods that solve larger combinatorial problems or physical-layer approaches that require detailed channel state information and precoding operations [2], [5], [14].

The resulting framework can be summarized as follows: the agent observes the macro-step state, generates K raw beam assignments sequentially, optionally applies the graph-based shield to obtain a conflict-free active set, evaluates service delivery and queue evolution, and updates the PPO policy during training. This design preserves the main contribution of the paper: sequential policy factorization improves learnability under inter-beam dependency, while inference-time graph-based shielding enforces deterministic scheduling-layer feasibility at deployment when strict graph-based interference avoidance is required.

V. Experiments and Discussion

This section evaluates the proposed shielded sequential reinforcement learning framework on a 63-cell physics-informed storm-aware synthetic benchmark. The evaluation addresses four questions: (i) whether sequential PPO improves scheduling efficiency compared with heuristic schedulers, (ii) whether conditional sequential factorization is more effective than simultaneous MultiDiscrete action prediction, (iii) whether tier-aware reward shaping improves hotspot service protection, and (iv) whether the inference-time graph-based shield guarantees conflict-free deployment under the adopted scheduling-layer interference graph.

The proposed environment should be interpreted as a physics-informed synthetic benchmark for scheduling evaluation, not as a full propagation simulator or meteorological digital twin. The storm process generates spatially structured, time-varying capacity degradation consistent with the qualitative behavior of rain-affected Ka-band satellite links. Therefore, the results should be interpreted as evidence of adaptive scheduling capability under controlled rain-induced nonstationarity, rather than as site-specific predictions for historical storm events.

A. Experimental Setting

All experiments are conducted on the 63-cell benchmark introduced in Section III. The system activates at most $K = 4$ beams per macro-step under full frequency reuse. All schedulers therefore operate under the same beam budget and the same adjacency-based conflict graph. The benchmark combines geographically asymmetric demand, diurnal traffic variation, queue dynamics, and synthetic storm-induced capacity degradation generated by the lightweight ITU-R-inspired attenuation model.

We evaluate four operating regimes: a no-rain scenario, a north-storm scenario, a south-storm scenario, and a reduced-load scenario. The no-rain regime represents nominal operation without rain-induced attenuation. The two storm regimes use representative deterministic synthetic storm tracks affecting northern and southern service regions, respectively. The reduced-load regime scales the offered traffic to approximately 75% of the nominal demand. The no-rain and reduced-load

TABLE I
MAIN SIMULATION AND ATTENUATION PARAMETERS.

Parameter	Setting
Service cells N	63
Beam budget K	4 per macro-step
Carrier frequency f_c	Ka-band: 20 GHz downlink / 30 GHz uplink
P.838 coefficients	$k = 0.07510$, $\alpha = 1.0991$ at 20 GHz V-pol
Rain height h_R	5.0 km
Station height	0.05 km
Link fade margin A_{margin}	50 dB
Peak storm intensity $I_{\max}$	$U(0.6, 1.0)$ per episode
Storm spread σ	$U(0.8°, 2.5°)$, approx. 90–280 km
Storm speed	$U(15, 60)$ km/h
Storm bearing	$U(225°, 315°)$
Macro-step duration	36 s
GEO longitude	108.0°E
Storm spawn points	12 coastal points with $\pm 0.3°$ jitter

scenarios are reported as mean $\pm$ standard deviation over five random seeds, whereas the storm scenarios are reported for representative deterministic synthetic storm tracks to illustrate scheduler behavior under spatially localized channel degradation.

Table I summarizes the main simulation settings. The storm-aware attenuation model assumes Ka-band operation, uses ITU-R P.838-3 specific attenuation coefficients, and maps attenuation to a normalized capacity factor through a conservative link fade margin. The storm-center translation speed is used only to move the synthetic storm footprint across the service region and should not be interpreted as wind speed or tropical cyclone category.

The PPO agent uses an actor–critic multilayer perceptron policy with ReLU activation. The action space at each internal sub-step is a discrete selection over the N service cells, and the observation follows the structured representation described in Section IV. The PPO training setup uses the clipped policy objective, value-function regression, entropy regularization, and generalized advantage estimation. The rollout horizon is synchronized with a complete simulated daily traffic cycle to reduce phase-dependent bias caused by the diurnal demand pattern. Unless otherwise stated, the learning rate, clipping threshold, discount factor, GAE parameter, entropy coefficient, batch size, and number of PPO epochs are kept fixed across PPO variants. The PPO-MultiDiscrete baseline is trained with a larger macro-step budget than the sequential PPO variants, giving the joint-action baseline additional training exposure and making the comparison conservative with respect to the proposed sequential factorization.

B. Compared Methods and Metrics

The compared schedulers are Random, Round Robin (RR), Greedy-Q, Greedy-QW, PPO-MD, PPO-NoTier, and PPO-Tier. Random uniformly selects beam targets, RR cyclically serves cells, Greedy-Q prioritizes the largest queue backlog, and Greedy-QW additionally considers waiting time. PPO-MD is a representative joint-action PPO baseline using simultaneous MultiDiscrete beam selection. PPO-NoTier is the sequential PPO variant without tier-aware reward shaping, while PPO-Tier is the full proposed method.

Whenever meaningful, each method is evaluated with and without the inference-time graph-based shield. In the tables, "S" denotes shielded execution, and "NS" denotes unshielded execution. This comparison separates the scheduling behavior learned by the policy itself from the deterministic feasibility guarantee provided by the shield under the adopted graph-based interference model.

We report average throughput (Thr), normalized queue depth (QD), packet loss, average conflict-graph violations (Interf), mean satisfaction ratio (MSR), satisfaction-based Jain's fairness index JFI_{sat}, and hotspot satisfaction for Hanoi and Ho Chi Minh City. For storm scenarios, we additionally report rain-wasted throughput (RWT), which measures the average service opportunity lost because beams are assigned to cells with rain-degraded effective capacity. Lower RWT indicates less waste on severely faded service opportunities. The metric definitions follow the formulations in Section III; all satisfaction values are reported in percentage form.

C. No-Rain Performance and Ablation Study

Table II reports the no-rain comparison. PPO-Tier achieves the strongest operating point in useful throughput, packet loss, and hotspot satisfaction. With shielded execution, PPO-Tier reaches 19.66 units per macro-step and zero graph-based interference. The unshielded PPO-Tier variant maintains almost the same throughput and only a very small interference count, indicating that the learned policy has internalized substantial spatial conflict awareness from reward feedback. In contrast, heuristic baselines degrade sharply when the shield is removed, especially Greedy-QW, whose performance collapses under unshielded execution.

Fairness metrics should be interpreted together with hotspot satisfaction, throughput, and packet loss. For example, Random with Shield obtains a high MSR because it distributes service broadly across many cells, but it fails to protect high-demand hotspots and incurs higher loss. Similarly, Greedy-QW with Shield achieves the highest JFI_{sat}, but its Hanoi and Ho Chi Minh City satisfaction values remain much lower than those of PPO-Tier. Thus, the proposed method should be interpreted as a better multi-metric operating point rather than as the maximizer of a single scalar metric.

Table III isolates the effects of action formulation and reward shaping. PPO-Tier and PPO-NoTier outperform PPO-MD in useful throughput and packet loss, confirming that conditional sequential factorization is better suited than simultaneous MultiDiscrete action prediction for this scheduling problem. This supports the motivation that exposing partial assignment history helps the policy learn inter-beam spatial dependency.

The comparison between PPO-Tier and PPO-NoTier shows the role of tier-aware reward shaping. PPO-NoTier achieves slightly higher demand-uniform fairness and MSR, but its hotspot satisfaction is substantially lower. PPO-Tier intentionally trades a small amount of uniformity for stronger protection of high-demand metropolitan cells, which is desirable in service-oriented satellite scheduling under geographically asymmetric demand.

TABLE II
OVERALL COMPARISON IN THE NO-RAIN SCENARIO. S: WITH SHIELD; NS: NO SHIELD.

Method	JFI_{sat}	Hn Sat	HCM Sat	Thr	QD	Interf	Loss	MSR
PPO-Tier (S)	0.9827 ± 0.0028	**94.06 ± 1.57**	**89.07 ± 1.86**	**19.66 ± 0.32**	41.16 ± 4.19	**0.0000 ± 0.0000**	**3.81 ± 0.98**	71.06 ± 1.58
PPO-Tier (NS)	0.9810 ± 0.0019	93.89 ± 1.90	88.62 ± 2.95	19.64 ± 0.32	41.16 ± 4.43	0.0042 ± 0.0011	3.84 ± 0.97	71.06 ± 1.35
Greedy-Q (S)	0.9668 ± 0.0011	77.81 ± 1.37	57.09 ± 3.15	16.97 ± 0.11	50.01 ± 4.36	**0.0000 ± 0.0000**	6.00 ± 0.46	61.82 ± 0.84
Greedy-Q (NS)	0.8411 ± 0.0232	43.47 ± 1.89	12.19 ± 2.38	9.42 ± 0.55	56.36 ± 1.99	1.4724 ± 0.1330	13.26 ± 0.57	38.30 ± 2.03
Greedy-QW (S)	**0.9918 ± 0.0020**	54.25 ± 4.89	53.94 ± 4.95	17.67 ± 0.13	49.07 ± 4.19	**0.0000 ± 0.0000**	5.33 ± 0.47	69.47 ± 1.61
Greedy-QW (NS)	0.4479 ± 0.0586	7.53 ± 0.97	0.04 ± 0.04	0.38 ± 0.04	56.85 ± 2.00	4.2462 ± 0.3873	22.27 ± 0.05	0.95 ± 0.06
RR (S)	0.8102 ± 0.0033	6.44 ± 0.21	1.38 ± 0.08	10.40 ± 0.05	46.21 ± 3.25	**0.0000 ± 0.0000**	12.71 ± 0.16	52.07 ± 0.29
RR (NS)	0.1708 ± 0.0001	3.45 ± 0.02	0.00 ± 0.00	1.19 ± 0.00	56.23 ± 2.14	3.3808 ± 0.0000	21.47 ± 0.02	6.41 ± 0.02
Rand (S)	0.9275 ± 0.0054	13.53 ± 0.71	11.93 ± 0.69	16.10 ± 0.21	**30.39 ± 2.78**	**0.0000 ± 0.0000**	8.29 ± 0.42	**77.73 ± 1.20**
Rand (NS)	0.9035 ± 0.0043	8.31 ± 1.03	9.71 ± 0.60	14.48 ± 0.19	34.21 ± 2.49	0.4184 ± 0.0099	9.63 ± 0.30	71.95 ± 1.14

TABLE III
ABLATION ON ACTION FORMULATION AND REWARD DESIGN IN THE NO-RAIN SCENARIO. MD: MULTIDISCRETE.

Method	JFI_{sat}	Hn Sat	HCM Sat	Thr	QD	Interf	Loss	MSR
PPO-Tier (S)	0.9827 ± 0.0028	**94.06 ± 1.57**	**89.07 ± 1.86**	**19.66 ± 0.32**	41.16 ± 4.19	**0.0000 ± 0.0000**	**3.81 ± 0.98**	71.06 ± 1.58
PPO-Tier (NS)	0.9810 ± 0.0019	93.89 ± 1.90	88.62 ± 2.95	19.64 ± 0.32	41.16 ± 4.43	0.0042 ± 0.0011	3.84 ± 0.97	71.06 ± 1.35
PPO-NoTier (S)	0.9894 ± 0.0012	73.99 ± 5.13	77.50 ± 5.40	19.63 ± 0.31	**40.56 ± 3.83**	**0.0000 ± 0.0000**	3.88 ± 0.95	**73.63 ± 1.63**
PPO-NoTier (NS)	**0.9886 ± 0.0023**	73.64 ± 5.48	78.08 ± 5.33	19.61 ± 0.33	40.73 ± 3.85	0.0047 ± 0.0012	3.93 ± 0.94	73.58 ± 1.86
PPO-MD (S)	0.9610 ± 0.0030	65.12 ± 5.97	86.68 ± 2.96	18.82 ± 0.22	38.38 ± 2.24	**0.0000 ± 0.0000**	5.00 ± 0.66	72.70 ± 1.82
PPO-MD (NS)	0.9583 ± 0.0038	65.63 ± 5.85	85.70 ± 2.35	18.44 ± 0.19	38.97 ± 2.45	0.0921 ± 0.0115	5.32 ± 0.69	71.29 ± 1.68

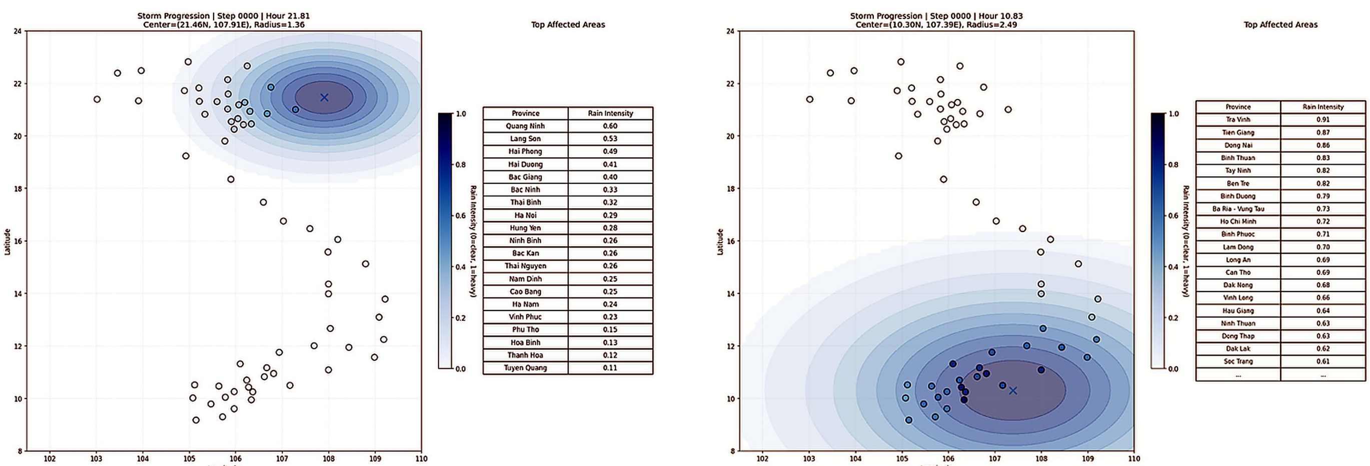

Fig. 2. Representative synthetic storm snapshots for the north-storm and south-storm scenarios.

D. Storm-Affected Scenarios

The north-storm and south-storm scenarios evaluate whether the scheduler can adapt when rain attenuation degrades the service opportunity of spatially localized regions. Figure 2 shows representative synthetic storm snapshots for the two cases. The maps are included only to illustrate the spatial structure of the controlled storm scenarios; they should not be interpreted as historical meteorological events.

Table IV summarizes the storm-scenario results using the most informative schedulers: the strongest heuristic baselines, the joint-action PPO-MD baseline, and the sequential PPO variants. In the north-storm scenario, PPO-Tier maintains strong throughput while protecting the southern hotspot when northern cells are rain-degraded. In the south-storm scenario, the trend reverses: Hanoi satisfaction remains high, while Ho Chi Minh City satisfaction decreases because the southern service region is more strongly attenuated. This behavior is consistent with rain-aware scheduling, where the policy jointly reacts to queue pressure, hotspot priority, and instantaneous channel opportunity.

Figure 3 visualizes RWT in the two storm scenarios. The RWT plots make the rain-aware behavior more visible than the final column of the result table. Although RWT should be interpreted together with throughput and packet loss, lower RWT generally indicates better avoidance of severely faded service opportunities.

E. Reduced-Load Behavior and Learned Policy Dynamics

In the reduced-load regime, the offered traffic is scaled to approximately 75% of the nominal demand. As expected, most methods improve when the system moves closer to the service-capacity boundary. PPO-Tier remains among the strongest schedulers in throughput, packet loss, hotspot satisfaction, and MSR, while preserving zero graph-based interference under

TABLE IV
COMPACT COMPARISON UNDER REPRESENTATIVE STORM-AFFECTED SCENARIOS.

Scenario	Method	Hn Sat	HCM Sat	Thr	Interf	Loss	RWT
North	PPO-Tier (S)	70.50	91.20	17.95	0.000	4.76	1.42
North	PPO-Tier (NS)	69.44	91.88	18.01	0.006	4.71	1.44
North	Greedy-Q (S)	65.09	43.27	14.34	0.000	8.30	2.14
North	Greedy-QW (S)	37.23	40.36	15.44	0.000	7.20	2.01
South	PPO-Tier (S)	96.23	18.83	16.56	0.000	7.91	2.11
South	PPO-Tier (NS)	96.16	20.84	16.54	0.018	7.91	2.09
South	Greedy-Q (S)	65.18	21.67	13.12	0.000	10.22	2.63
South	Greedy-QW (S)	35.64	20.30	13.43	0.000	10.10	3.50

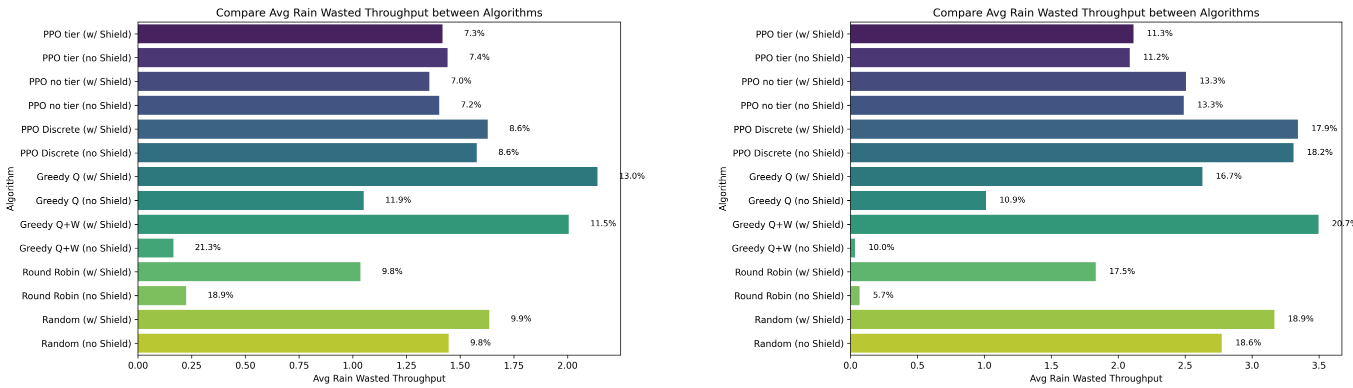

Fig. 3. Rain-wasted throughput comparison in the north-storm and south-storm scenarios.

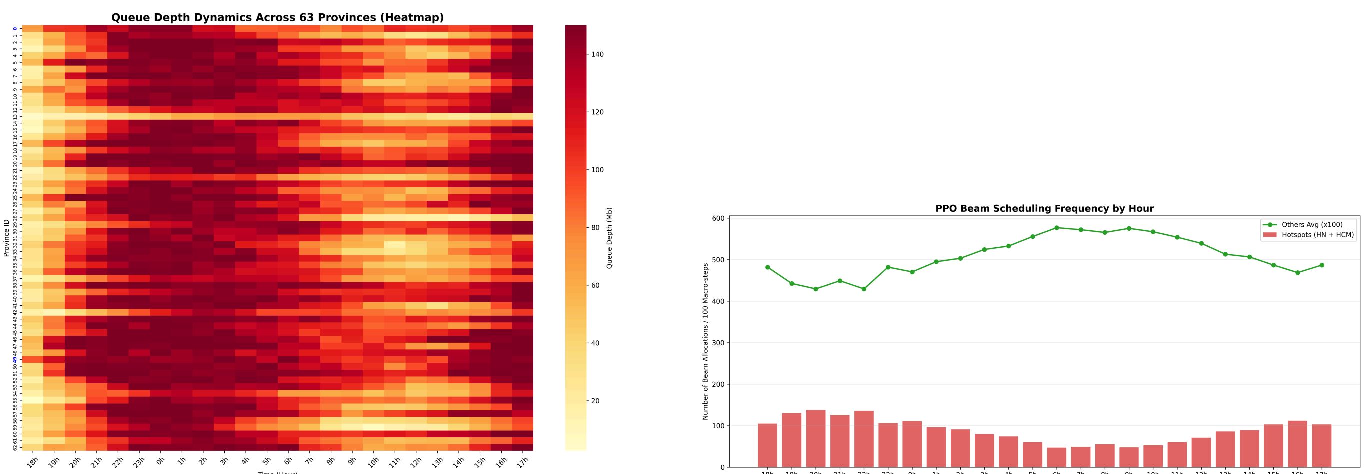

Fig. 4. Learned scheduling behavior over time: daily service heatmap and hourly service pattern for major hotspot cells.

shielded execution. The small performance gap between PPO-Tier with and without the shield again suggests that the learned policy itself has acquired interference-aware scheduling behavior. For space reasons, the full reduced-load table is omitted; the main role of this scenario is to confirm that the proposed method does not rely solely on persistent overload.

Beyond aggregate metrics, Fig. 4 illustrates the learned scheduling behavior over time. The daily service heatmap shows that the policy does not follow a static allocation pattern. Instead, it redistributes beam service across cells according to queue pressure, time-of-day demand variation, and channel conditions. The hotspot service profile further shows that the learned scheduler responds to diurnal demand changes and increases service to major hotspots during high-demand periods.

Figure 5 shows the training reward curve of the PPO scheduler. The increasing trend indicates that the policy gradually learns more effective beam assignment patterns as training progresses. However, reward curves should be interpreted together with evaluation metrics, because the final objective is not only to maximize scalar reward but also to maintain throughput, reduce loss, protect hotspots, and satisfy deployment-time feasibility.

F. Discussion

The experimental results lead to five observations. First, conditional sequential factorization improves the learnability of multibeam scheduling. Compared with simultaneous Mul-

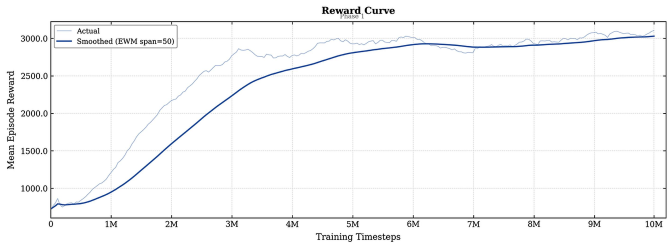

Fig. 5. Training reward curve of the proposed PPO scheduler.

tiDiscrete PPO, the sequential policy achieves higher useful throughput and lower packet loss, while providing stronger hotspot satisfaction. This supports the hypothesis that exposing the partial beam assignment to the policy helps resolve inter-beam dependency and structural credit assignment.

Second, shielded and unshielded comparisons show that the learned policy itself acquires spatial conflict awareness. For PPO-Tier, removing the shield causes only marginal interference, whereas heuristic schedulers degrade substantially without shielded execution. Thus, the shield should be viewed as a deterministic deployment-time feasibility layer rather than a post-processing mechanism that creates the scheduling intelligence.

Third, tier-aware reward shaping primarily improves service protection for high-demand cells. The gain is most visible in Hanoi and Ho Chi Minh City satisfaction, while aggregate throughput remains comparable to the non-tiered variant. This behavior is desirable in geographically asymmetric service regions, where equal resource allocation does not imply equal demand satisfaction.

Fourth, the storm-affected scenarios demonstrate that the proposed scheduler remains effective under spatially structured channel degradation. The policy maintains higher throughput and lower loss than queue-based heuristics, indicating that it uses rain-dependent capacity information rather than serving only the largest queues.

Fifth, the reduced-load scenario shows that the method does not rely solely on persistent overload. When the offered traffic is reduced, PPO-Tier continues to maintain high satisfaction and zero shielded interference, suggesting that the learned policy remains useful across different operating regimes.

VI. Conclusion

This paper presented a shielded sequential reinforcement learning framework for rain-aware multibeam satellite scheduling. The proposed method reformulates each physical scheduling interval as a sequence of conditional beam-selection decisions, allowing the policy to condition each decision on the partial beam assignment already constructed within the same macro-step. This design exposes inter-beam spatial dependency to the learning agent while preserving simultaneous beam activation at the scheduling interval. An inference-time graph-based shield is further introduced as a deployment-time feasibility layer to project the raw beam sequence onto a conflict-free active set under the adopted scheduling-layer interference graph.

Experiments on a 63-cell physics-informed storm-aware benchmark show that the proposed PPO-Tier scheduler achieves a stronger multi-metric operating point than heuristic and MultiDiscrete PPO baselines. The results confirm that conditional sequential factorization improves useful throughput and packet loss compared with simultaneous joint-action prediction, while tier-aware reward shaping improves service protection for high-demand hotspot cells. The comparison between shielded and unshielded execution further indicates that the learned policy acquires substantial spatial conflict awareness, with the shield serving primarily as a deterministic graph-based feasibility layer rather than the source of scheduling intelligence.

These findings suggest that combining conditional policy generation with inference-time graph-based feasibility enforcement is a promising design principle for cognitive multibeam satellite scheduling under geographically asymmetric demand, dynamic traffic, and rain-induced channel degradation. Future work will extend the current physics-informed synthetic benchmark toward historical storm tracks, time-varying rain fields, SINR-aware interference modeling, adaptive coding and modulation, and broader transfer evaluation across GEO and NGSO satellite settings.

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix
Proof of the Zonklar Equations

Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

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Biography Section

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